Lalith Chandra Malladi

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Summary

RTL design engineer with experience in reconfigurable LLM inference hardware, FPGA ML accelerators, and DDR memory interfaces. Seeking ASIC/FPGA design roles in computer architecture.

Experience

Vicharak Computers: ASIC Design Engineer Dec 2025 - Present

LLM Acceleration. Surat, Gujarat, India

- RTL design and simulation of reconfigurable activation NoC in LLM inference (SystemVerilog, C, Haskell).
- Analysis and redesign of configuration memory for coarse-grained architectures for reduced area and latency in partial reconfiguration.

Vicharak Computers: FPGA Developer Intern Jun 2025 - Dec 2025

Gati: an Edge ML Accelerator. Surat, Gujarat, India

- Built and verified parameterized fixed-latency compute blocks in Verilog (Non-linear activation functions) for Vaaman board (Efinix T120 FPGA).
- Debugged LPDDR3 and DDR3L memory interfaces on custom PCBs (Frequency and timing parameters), performed DDR calibration and training.

VLSI Research Intern: National Institute of Technology, Delhi May 2024 - Jul 2024

Self-proposed project. Delhi, India

- Architected a 32-bit approximate image multiplier with Vivado (Verilog for Artix-7 FPGA), achieving 30% lower power consumption in image processing workloads.
- Completed physical design and functional verification of the multiplier as ASIC for 90nm and 180nm technologies (Cadence Genus, Innovus, NCSim), ensuring design met PPA constraints.

Education

Rajiv Gandhi University of Knowledge Technologies, Basar, TG Jun 2021 - May 2025

B.Tech., Electronics and Communication Engineering 7.77 GPA

Rajiv Gandhi University of Knowledge Technologies, Basar, TG Jun 2019 - May 2021

Pre-University Course (PUC), Class XI & XII equivalent 8.86 GPA

Personal Projects

BI-GPU: Open-Source GPU (In Development) Jul 2026 - Present

Designing a basic GPU with SystemVerilog.

- Parameterized shader core with register files, program counter and custom ISA decoder.
- ISA simulation and testing with Verilator and Python. Public release planned after memory hierarchy (cache, MMU).

Academic Projects

Advanced AXI FireFly Architecture for SNN Acceleration Sep 2024 - May 2025

Enhanced a high-performance VLSI architecture for spiking neural network acceleration (Vivado, Python, SpinalHDL).

- Integrated approximation units into the FireFly accelerator pipeline to improve throughput and reduce area.
- Generated synthesizable Verilog from **SpinalHDL (Scala)** and custom blocks integrated at the RTL level. Verification done with cocotb.

License Plate Networking (V2V Communication) Sep 2023 - Jan 2024

Proposed and led the design of a vehicle-to-vehicle communication system using license plates as unique identifiers.

- Designed the license plate detection pipeline using a custom-trained **YOLOv8** model, achieving 97% detection accuracy (Python, Google Colab).
- Used **EasyOCR** for text extraction and built a cropping module for clean text generation.
- Delivered the complete detection and recognition module as the core component for further system integration.

Approximate MAC Unit Feb 2024 - May 2024

- Designed an approximate Multiply-Accumulate unit using an 8-bit truncation adder and 11-bit approximate accumulator.
- Verified functionality and power savings using Icarus Verilog (iverilog), Vivado, and GTKWave.

Co-curriculars

Secretary: Entrepreneurship Cell, RGUKT Basar Sep 2022 - Jan 2025

- Progressed from PR Manager to Corporate Relations Manager and elected as the Secretary of the university's Entrepreneurship Cell.
- Organized an overnight hackathon on campus with 36 participating teams (250+ students) and judged 20+ startup ideas.
- Mentored and supported 3 student startups to advance to the zonal rounds of IIT Bombay's Eureka! competition.

Technical & Creative Team Member: Department of ECE, RGUKT Basar Sep 2022 - May 2025

- Mentored multiple student projects showcased at the university's AntahPragnya 2025 Technical Festival.
- Led graphic design and on-stage technical operations for major departmental events.

Activities and Achievements

- Invited by Telangana Innovation Cell (TGIC) to attend and was officially sponsored for **Startup Mahakumbh 2024**.
- Moderated the Student Dialogue Panel at Kakatiya Sandbox's 6th and 7th Development Dialogue.
- Secured first runner-up at the National Entrepreneurship Challenge (NEC) 2024 and placed in the **Top 9 at NEC 2023**.

Technical Skills

Languages: Verilog, SystemVerilog, Python, C, Haskell, Scala, Bash
EDA & Open-Source Tools: Efinity IDE, Vivado, Verilator, IVerilog, GTKWave, cocotb, Cadence Genus & Innovus
Hardware & Systems: Efinix FPGAs, reconfigurable SBCs, DDR3L/LPDDR3, AXI, Linux OS